

Program Design for Maximal Hypertrophy

This chapter builds on the information from previous chapters to explore the practical application of the science of hypertrophy training. Considerations for exercise selection are discussed from a biomechanical standpoint with a focus on how movements can be synergistically varied to ensure complete muscular development. A discussion of program design follows, detailing the nuances of manipulating program variables over the course of a periodized training cycle to maximize the hypertrophic response by proper management of stimulus and fatigue. Numerous examples are provided throughout the chapter to illustrate the practical application of relevant concepts. It is important to understand that these examples represent the art of program design and are for illustrative purposes only. While paying proper attention to underlying scientific principles, lifters should harness their personal experience in conjunction with their own needs and abilities to formulate a strategic plan. This is the essence of an evidence-based approach to training.

Biomechanics

Biomechanics is the study of how internal and external forces affect the living body; particular attention is given to the musculoskeletal system. A variety of biomechanical factors must be taken into account when choosing exercises for a hypertrophy-oriented program. These include the length–tension relationship, training angle, plane of movement, spacing of

hands and feet, and exercise type, which are addressed in this section. The ensuing section, Exercise Selection Strategies, explores how to apply these factors to resistance training program design to maximize hypertrophy.

Length–Tension Relationship

The capacity of a muscle fiber to produce force is predicated on the position of the actin and myosin filaments in its sarcomeres. This phenomenon, known as the *length–tension relationship* (figure 8.1), can be harnessed to target muscles or portions of them by making them more or less active during exercise. Optimal force-producing capacity is often said to take place at approximately resting length, whereby the overlap of actin and myosin filaments is maximized, thus facilitating optimal crossbridge formation. However, working a muscle at 125% to 140% of resting length may confer even greater benefits on force output because the stretching of sarcomeres brings the myofilaments together and enhances calcium sensitivity; it is hypothesized that the greater potential for crossbridge attachment from the

KEY POINT

Length–tension relationship, training angle, plane of movement, spacing of hands and feet, and exercise type can all be carefully manipulated in program design to maximize hypertrophy.

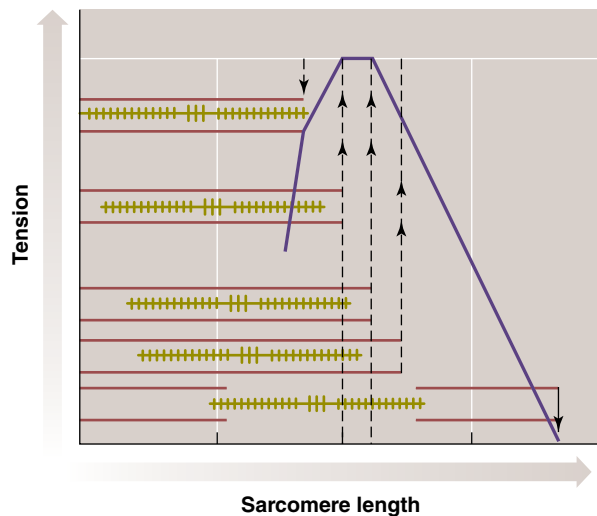


FIGURE 8.1 The length–tension relationship.

closer proximity of myofilaments and heightened calcium affinity overcomes the detriment of fewer myosin heads in the region of overlap (84).

Two primary strategies can be employed to take advantage of the length–tension relationship from an exercise selection standpoint: active insufficiency and passive tension. *Active insufficiency* refers to when a two-joint muscle is shortened at one joint while a muscular action is initiated at the other joint. Because a muscle loses the ability to shorten when its attachments are close together, it is in a functionally disadvantageous position on the length–tension curve, resulting in a diminished capacity to produce force. For example, when the shoulder is in the flexed position during performance of the biceps curl, the biceps brachii’s origin at the scapula and insertions below the elbow are brought closer together, and the bicep’s ability to produce force is therefore limited. Alternatively, *passive tension* refers to when a two-joint muscle is elongated at one joint while carrying out dynamic movement at the other joint. This produces a favorable length–tension relationship, enhancing the muscle’s ability to produce force. For example, the long head of the triceps brachii crosses both the shoulder and elbow joints, carrying out shoulder flexion and elbow extension at these joints, respectively. Because the muscle is shortened during shoulder exten-

sion, it is lengthened during shoulder flexion. Thus, performing an exercise in which the shoulder joint is flexed (such as the overhead triceps extension) places the muscle in a position of stretch while carrying out its action at the elbow and consequently allows for greater force production.

It should be noted that viewing the length–tension relationship in isolation somewhat simplifies the complexity of in vivo kinetics. A variety of factors affect the functional force–length range, including the absolute muscle length, the number of sarcomeres, tendon length and stiffness, the length of the moment arm, and the range of motion of the acting joint or joints (84). In addition, changes in both active forces (from the myofilaments) and passive forces (from elastic components such as titin, fascia, and tendon) take place throughout a joint’s range of motion (84), which in turn may alter the hypertrophic stimulus. Nevertheless, using the concepts of active insufficiency and passive tension to target different muscles is an uncomplicated and viable strategy for guiding exercise selection.

Training Angle

Muscle fibers contract optimally when placed in direct opposition to gravity along the direction of the fiber. Changing the angle of training at which a muscle is worked best targets the full spectrum of its fibers, allowing for more symmetrical muscular development. Thus, the orientation of fibers in a given muscle must be considered when selecting exercises. For example, performing the lateral raise with the shoulder joint externally rotated positions the anterior deltoid to directly oppose gravity; to target the middle deltoid head requires performing the movement in internal shoulder rotation, which orients these fibers to carry out the majority of work.

Movement Plane

The human body is designed to move in three-dimensional space. To account for this capability, the body can be segmented into three anatomical planes (figure 8.2): *sagittal*, which divides the body into left and right

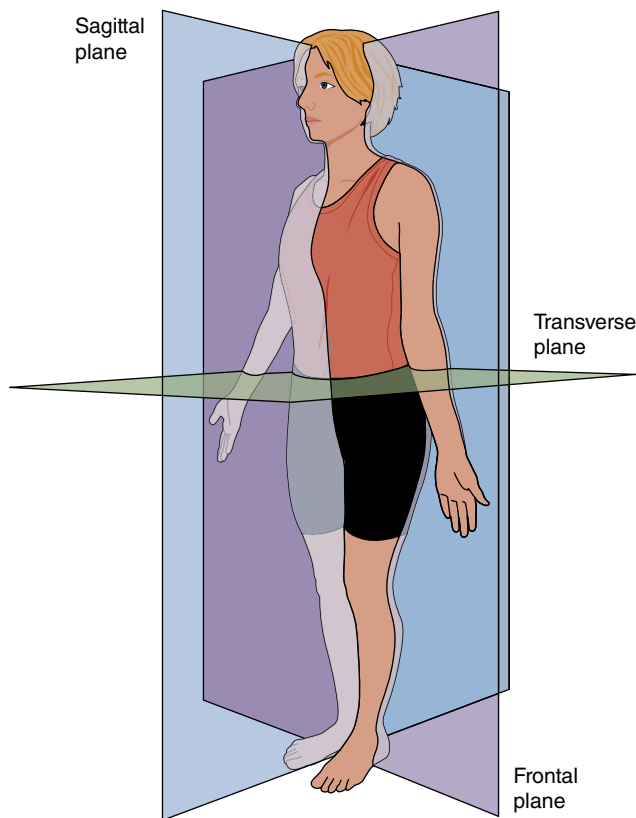


FIGURE 8.2 The planes of movement.

halves and encompasses flexion and extension; *frontal* (i.e., coronal), which divides the body into front and back sections and includes abduction, adduction, elevation, depression, inversion, eversion, and lateral flexion; and *transverse*, which divides the body into top and bottom portions and includes horizontal adduction, horizontal abduction, rotation, pronation, and supination. Note that although these planes are rigidly defined, diagonal movement in all planes is possible depending on the task requirement and individual mobility.

To carry out movement efficiently and effectively, the musculoskeletal system summons muscles based on the directional requirements of the task. As such, muscular activation changes based on the plane of movement in which the body is worked. The application of training in various planes to maximize muscular development depends on the degrees of freedom of the joint. Joints that have multiple

degrees of freedom (e.g., ball-and-socket joints) can benefit from multiplanar training, whereas those with a single degree of freedom (e.g., hinge joints) do not.

Spacing of Hands and Feet

The positioning of the extremities can alter muscle activation patterns. The orientation of fibers within a given muscle ultimately dictates the extent to which changes in hand and foot spacing influence activation. The effects of such alterations tend to be rather subtle, but nevertheless can be sufficient to promote meaningful differences in muscle development.

Exercise Type

Multi-joint exercises involve the dynamic activation of numerous muscles while statically engaging many stabilizers. Moreover, because loading is dispersed over multiple joints and muscles, heavy weights can be employed to maximize mechanical tension without creating undue joint stress. Hence, multi-joint exercises provide an effective means to train the entire body efficiently. However, they are limited because some muscles make a greater contribution to movement than others. Single-joint exercises afford the ability to directly target individual muscles and elicit unique neuromuscular activation patterns that enhance overall muscular development (7). The torque-angle curves of single-joint exercises must be taken into account in program design. Contreras and colleagues (33) employed biomechanical modeling to propose a three-part torque-angle classification system for single-joint exercises:

1. Long-length accentuated force exercises create maximal torque while the prime movers are stretched (e.g., chest fly; figure 8.3a).
2. Short-length accentuated force exercises create maximal torque while the prime movers are shortened (e.g., hip thrust; figure 8.3b).
3. Midlength accentuated force exercises create maximal torque while the prime movers are between the extremes (e.g., 45° back extension; figure 8.3c).



FIGURE 8.3 Exercises typifying a torque-angle classification system for single-joint exercises: (a) chest fly—maximal torque while the prime movers are stretched, (b) hip thrust—maximal torque while the prime movers are shortened, and (c) 45° back extension—maximal torque while the prime movers are between the extremes.

PRACTICAL APPLICATIONS

ATTENTIONAL FOCUS AND MUSCLE HYPERTROPHY

Attentional focus is a well-recognized aspect of motor learning and its use has important implications for muscular hypertrophy. Operationally defined from a resistance training standpoint, *attentional focus* refers to what a person thinks about during each repetition. Two primary types of attentional focus have been recognized in the literature: internal and external. An *internal focus* involves thinking about bodily movements during performance, whereas an *external focus* involves thinking about the outcomes of movements.

The majority of research supports adopting an external focus of attention when carrying out performance-oriented tasks. A recent comprehensive review of the literature found superior effects from using an external versus an internal focus in more than 90% of studies that examined performance-oriented outcomes (151). The performance-based superiority of an external focus during resistance training is thought to be due to an enhanced economy of movement associated with greater force production and reduced muscular activity compared to an internal focus (constrained action hypothesis) (86). It is important to note, however, that improvements in performance-related measures do not necessarily equate to maximal increases in muscle hypertrophy. A case can be made that an internal focus is a better approach when the goal is to maximize muscle development.

Employing a hypertrophy-oriented internal focus of attention is consistent with the long-standing bodybuilding axiom of developing a mind–muscle connection. Simply stated, this strategy involves visualizing the target muscle during the course of a lift and willfully directing neural drive to that muscle. When properly executed, the approach theoretically allows for increased stimulation of the target muscle while reducing the involvement of other synergists.

Indirect evidence lends support to a hypertrophic benefit when using an internal focus. Numerous studies have found that activation of a given muscle is enhanced by using an internal focus of attention. Snyder and Leech (127) demonstrated that subjects were able to significantly increase electromyography (EMG) activity in

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Attentional Focus and Muscle Hypertrophy *(continued)*

the latissimus dorsi by directing their focus to this muscle during the lat pulldown exercise. A follow-up study by the same lab showed that the pectoralis major and triceps could be individually targeted after subjects were instructed to visualize those muscles during performance of the bench press at 50% of 1RM (128). Interestingly, the magnitude of the effect was substantially reduced when the load was increased to 80% of 1RM. This may be due to increased force demands when training with heavier loads, thereby altering the ability to focus on the muscle being worked in favor of simply lifting the load. The implication is that the hypertrophy-related benefits of using an internal focus may be attenuated or annulled when training with very heavy loads. That said, the ability to increase muscle activation through an internal focus has been shown in other muscles as well, including the abdominals (20, 35, 68), gluteus maximus (81), and elbow flexors (86, 137). The findings provide a strong rationale for using an internal focus to target a given muscle.

The logical question is whether increasing activation of a muscle translates into greater muscle growth. Although research on the topic remains limited, some evidence suggests that this is indeed the case. Wakahara and colleagues (138) carried out a two-part experiment to investigate the topic. In the first part of the experiment, muscle activation was assessed by T2-weighted magnetic resonance imaging during 5 sets of 8 repetitions of the lying triceps extension in 12 untrained men. The results showed that activation of the triceps brachii was significantly higher in the proximal and middle aspects of the muscle versus the distal portion. In the second part of the study, 12 additional subjects performed the same routine used in part 1 of the study for 3 days per week over 12 weeks. At the study's conclusion, increases in muscle cross-sectional area corresponded to the specific regions most activated during exercise performance. A follow-up study by the same lab reported similar findings using alternative exercises for the triceps brachii (139). Although subjects were not employing a specific attentional focus, the findings nevertheless indicate that greater activation can translate into greater increases in muscle mass.

Our lab conducted the only study to date to directly investigate the effects of attentional focus on muscle hypertrophy (120). Thirty untrained men were randomized to perform biceps curls and leg extensions using either an internal focus (i.e., focus on the muscle) or an external focus (i.e., focus on the outcome of the lift). Both groups performed 4 sets of 8 to 12 repetitions per exercise, with training carried out 3 days per week. After 8 weeks, the internal focus group showed significantly greater increases in elbow flexor thickness compared to those adopting an external focus (12.4% vs. 6.9%, respectively). Alternatively, both groups achieved similar increases in quadriceps growth. Although speculative, discrepancies between muscle groups may be attributed to the fact that most people find it easier to develop a mind–muscle connection in the upper extremities because the arms are used for actions that require dexterity and thus more brain-coordinated fine motor control. On the other hand, the lower body is used primarily for ambulation, and these gross movement patterns require less conscious thought to carry out.

In totality, the findings of increased muscle activation combined with those showing site-specific hypertrophy in the region of activation seem to suggest that an internal attentional focus is the best approach for maximizing muscle development. The only study to date that directly investigated the topic provides further support for such an approach. Although many gym-derived tenets of bodybuilding are of questionable practice, claims of the hypertrophic benefit of developing a mind–muscle connection and employing it during exercise performance seem to have merit.

Exercise Selection Strategies

Selecting the appropriate exercises is an important factor for maximizing whole-body muscle hypertrophy. For example, certain muscles have multiple attachments that improve leverage for movement patterns. Moreover, myofibers often are subdivided into neuromuscular compartments, each of which is innervated by its own nerve branch (144, 148). Functionally independent muscle segments facilitate the central nervous system's ability to fine-tune human movement for optimum efficiency during complex motor tasks (143). Importantly, these inter- and intramuscular architectural variances reinforce the need to adopt a multiplanar, multiangled approach

KEY POINT

Maximal hypertrophy can be best achieved by systematically varying the exercises performed and fully working all aspects of the targeted musculature, varying the angles and planes involved, and using both multi-joint and single-joint exercises.

to hypertrophy-oriented training using a variety of exercises. Maximal hypertrophy can be achieved only by systematically varying the exercise performed and fully working all aspects of the targeted musculature. This section explains how to employ these strategies to maximize hypertrophy in each of the major muscle groups.

PRACTICAL APPLICATIONS

HOW TO CALCULATE VOLUME IN MULTI- VERSUS SINGLE-JOINT EXERCISES

Resistance training volume recommendations for hypertrophy are generally based on meta-analytic data that endeavor to quantify the number of sets performed per muscle group per week (i.e., set volume). However, a conundrum arises when deciding how to account for volume during multi-joint versus single-joint exercise. In these movements, working muscles can act as agonists (a prime mover in carrying out the exercise), synergists (a secondary mover that contracts simultaneously with the prime mover in performance), or stabilizers that contract isometrically to maintain postural stability.

A recent meta-analysis on the topic gave equal weight to both agonists and synergists when calculating volume during multi-joint exercise (119). Thus, for determining hypertrophy of the triceps brachii, a set of the bench press (multi-joint exercise) and triceps pushdown (single-joint exercise) were counted on a 1:1 basis. The same principle applied for the biceps brachii during lat pulldowns (multi-joint exercise) and arm curls (multi-joint exercise), and the quadriceps during the leg press (multi-joint exercise) and leg extension (multi-joint exercise). The approach was justified by findings of a recent review that concluded the performance of multi-joint and single-joint exercises produces similar increases in muscle size (54).

However, while it is clear that multi-joint exercise can promote significant hypertrophy in the synergists, the extent of their stimulation during these movements remains questionable. Muscle activation is influenced by a variety of biomechanical factors, including the length–tension relationship, muscle moment arms, and motor abundance (i.e., the body's attempt to determine a unique solution to efficiently perform a complex motor task). The interplay of these variables is complex, varying between exercises and, to some extent, individuals. However, taking into account

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How to Calculate Volume in Multi- Versus Single-Joint Exercises *(continued)*

the ability to alter these various biomechanical factors to more favorably work a given muscle or segment of a muscle (see the biomechanics section in this chapter for a more detailed discussion), it seems logical that single-joint exercise can potentially elicit greater hypertrophy for certain muscles compared to multi-joint movements, at least in certain exercises and under certain conditions.

Numerous electromyographic (EMG) studies report differences in muscle activation between multi-joint and single-joint exercise. For example, single-joint exercises targeting the hamstrings (e.g., leg curl, stiff-leg deadlift) display significantly greater EMG amplitudes than multi-joint lower-body exercise (e.g., squat, leg press) (5, 150). With respect to the quadriceps, studies show higher EMG amplitudes for the rectus femoris during single-joint knee extension exercise versus multi-joint exercises such as the barbell squat and leg press (5, 44). Discrepancies in muscle activation between muscles during multi-joint exercises have been shown for the upper-body musculature and provide further insights on the topic. Activation of the pectoralis major is approximately twice as great as that of the triceps brachii during performance of the bench press (27, 108), and EMG amplitude of the biceps brachii is lower than that of the latissimus dorsi in the lat pulldown and seated row exercises (79, 82). It is important to note that although some evidence indicates a correlation between muscle activation and increases in hypertrophy (138-140), causality cannot be inferred from correlational data, and the efficacy of EMG in predicting future hypertrophic changes remains undetermined.

Longitudinal research on the topic remains somewhat equivocal; some studies demonstrate a potential superiority of single-joint compared to multi-joint exercise (12, 13, 85), and others show no apparent differences (11, 14, 37, 52, 53). Further confounding matters, many of the studies measured muscle mass by the circumference method, which displays limited ability to predict hypertrophic changes. That said, the collective body of evidence seems to suggest that single-joint exercises provide an added benefit to maximizing muscle growth, as discussed in chapter 4. A particular benefit appears relevant to targeting individual heads of a given muscle (unpublished findings). It should be pointed out that current research is specific to the elbow flexors and extensors; the lack of studies comparing the effects of single-joint and multi-joint exercises on lower-body muscle development precludes the ability to draw strong inferences about this musculature.

As noted in a recent review (121), practitioners are best served by viewing set and volume prescription for single- and multi-joint exercises on a 1:1 basis, and then using logical rationale and personal expertise to guide exercise program design. When customizing exercise prescription, both the biomechanical and physiological aspects of an exercise should be taken into account in accordance with applied anatomy of the target muscle, consistent with the needs and abilities of the individual.

Back

The back muscles benefit from being trained in all three planes of movement. The frontal and sagittal planes, in particular, should be exploited to optimize muscular development. The latissimus dorsi (lats) are maximally stimulated by humeral adduction carried out in the frontal plane. The pull-up and lat pulldown exercises using a pronated grip are excellent for targeting the lats (82, 154). Grip widths in these movements show minor differences in muscle activation, but varying these positions from shoulder-width to twice shoulder-width distance may help to fully stimulate the musculature (6).

The midback muscles (middle trapezius and rhomboids) are best targeted using sagittal plane exercises (e.g., bent-over row and seated row). A neutral grip reduces biceps brachii activation, which seemingly allows the back musculature to carry out a greater amount of work. Despite a logical basis, there does not appear to be an added benefit to actively retracting the scapulae during rowing movements (79).

Single-joint shoulder extension exercises in the sagittal plane such as the pullover are often recommended for lat development. There is evidence that muscle activation in the pullover significantly favors the pectoralis major more than the lats, and the level of activation depends on the external force lever arm produced (87). However, the pullover exerts a great stretch in the lats at the start position, which may accentuate growth via increased myodamage or perhaps other factors related to stressing a muscle at long lengths under load. Therefore, the pullover, with a focus on accentuating the beginning phase of the movement, can be a useful addition to a hypertrophy-oriented routine.

Chest

The pectoralis major is maximally activated in the transverse plane using horizontal adduction movements. Both multi-joint exercises (horizontal, incline, and decline bench press) and single-joint exercises (horizontal, incline, and decline chest fly) are viable choices to develop the chest musculature. Pressing

movements allow for the use of heavier loads, and the chest fly provides greater isolation of the target muscles at the relative exclusion of assistors (67). A combination of both types of exercises therefore conceivably maximizes the hypertrophic response, although evidence for this hypothesis is lacking.

The pectorals can benefit from the use of a variety of training angles. The sternal head is best targeted during flat supine exercises (figure 8.4a) and decline exercises (figure 8.4b) (55), whereas the clavicular head is more aligned with gravitational forces when the torso is inclined at an angle of 30° to 45° (figure 8.4c) (76, 136). Hand spacing also influences pectoral muscle activation. A narrow grip elicits greater activation of the clavicular head (15). This is likely due to the fact that a narrow grip brings the elbows close to the torso, which makes the exercise a sagittal plane shoulder flexion movement. Single-joint overhead shoulder extension exercises such as the dumbbell pullover (figure 8.4d) substantially activate the sternal head of the pectoralis major (87), making it a viable addition to a comprehensive training program.

Torque angle during chest training also must be considered with respect to the modality of exercise. Barbell and dumbbell exercises heavily load the pectoralis major in the early phase of movement, but the musculature becomes increasingly unloaded at the finish position. Conversely, cable pulleys and many machines allow for a more constant muscular tension throughout the range of motion (ROM), which enhances muscular stimulation and metabolic stress in the pectorals. Thus, employing a variety of modalities would seemingly benefit hypertrophic adaptations. The addition of bands or chains can help to balance out the strength curve in free weight exercises, potentially enhancing their effectiveness (24, 51).

Shoulder

The deltoids are partitioned into three distinct heads that function in each of the anatomical planes: The anterior head is a shoulder flexor and thus is targeted with sagittal plane movements (e.g., front raise), the middle head is an abductor and thus is targeted with frontal plane movements (e.g., lateral raise), and the poste-



FIGURE 8.4 Exercises that target the pectorals from a variety of training angles: (a) flat bench press, (b) decline bench press, (c) incline bench press, and (d) dumbbell pullover.

rior head is a horizontal abductor and thus is targeted with transverse plane movements (e.g., reverse shoulder fly, bent-over lateral raise) (19). Research shows that the individual heads are further subdivided into at least seven separate muscle segments, each with the potential to be independently coordinated by the central nervous system (143); however, the training-related implications of these segments are not clear.

Shoulder rotation also must be considered when working the deltoids. The shoulder press, a frontal plane exercise, is generally thought to target the middle head of the deltoid. However, because the shoulder joint is externally rotated during performance, the anterior head is placed in a position to directly oppose gravity and thereby receives the majority of stimulation; the middle and posterior heads are substantially less active (19). Internal shoulder rotation is needed to place the middle head in a position

to directly oppose gravity, which is naturally accomplished in the wide-grip upright row (90, 113). Similarly, an internally rotated shoulder (i.e., pinky up) should be maintained during the lateral raise for optimal stimulation of the middle deltoid. An externally rotated shoulder position during horizontal abduction exercise was shown to best target the posterior deltoid (115), although personal preference seems to be most important given the fairly large inter-individual responses noted between subjects.

Upper Arm

The elbow is a hinge joint and thus moves in only one plane (sagittal). The muscles acting at the elbow are heavily involved in multi-joint upper-body exercises such as presses, pull-ups, and rows. However, both the elbow flexors and the elbow extensors contain *biarticular* (crossing two joints) muscles. The length–tension

relationship of these muscles is therefore sub-optimal during multi-joint exercises. Accordingly, targeted single-joint exercises afford the potential for stronger muscular contractions and thus greater growth.

With respect to the elbow flexors, the biceps brachii crosses both the shoulder and elbow joints. The long head, in particular, acts as a shoulder flexor (80), which makes it maximally active in exercises in which the humerus is extended behind the body (e.g., incline biceps curl; figure 8.5*a*). The long head also functions as a humeral abductor. The short head, therefore, can be targeted by performing exercises in which the humerus is abducted to 90° because the long head is somewhat actively insufficient in this position (59). Considering that the biceps are powerful radioulnar supinators, performing exercises with the hands neutral (e.g., hammer curl; figure 8.5*b*) or pronated (e.g., reverse curl; figure 8.5*c*) renders the biceps actively insufficient, thereby progressively increasing the work of the brachioradialis and brachialis muscles, respectively.

With respect to the elbow extensors, the long head of the triceps brachii has an optimal length–tension relationship when the shoulder is flexed to about 180° (77), meaning that this aspect of the musculature is most active during exercises in which the humerus is held overhead (e.g., overhead triceps extension).

Conversely, the medial and lateral heads are more active during movements such as the triceps pushdown, in which the humerus is held at the sides (139). This renders the long head less active so that the other heads carry out a greater amount of work. The applied theory is somewhat supported by the findings of Stasinaki and colleagues (130), who compared triceps training at a long muscle length (overhead triceps extension) versus at a short muscle length (triceps pushdowns) in untrained subjects. Although no statistically significant post-study differences were observed in growth of the long head of the triceps after 6 weeks of training, gains favored the group that employed the overhead extension for both muscle thickness (15% vs. 10%), and cross-sectional area (16% to 25% vs. 14% to 17%). Two separate experiments by Wakahara and colleagues (138, 139) lend further support to the concept. In one study (139), 12 weeks of training with the close-grip bench press elicited significantly greater hypertrophy in the midportion of the triceps (corresponding to the medial and lateral heads) compared to the proximal portion (corresponding to the long head of the triceps). In the other study, greater hypertrophy was observed in the proximal portion (long head) compared to the distal and midpoints following 12 weeks of performing the lying triceps extension (138).



FIGURE 8.5 Exercises to target the elbow flexors: (a) incline biceps curl, (b) hammer curl, and (c) reverse curl.

Hip

The gluteals make up the primary muscle group of the hip and include the gluteus maximus, gluteus medius, and gluteus minimus. The gluteals function in all three planes of movement, but particularly in the transverse and frontal planes. Sagittal plane multi-joint exercises for the lower body, such as the squat, lunge, and leg press, heavily involve the gluteus maximus. A wide stance increases activation of the gluteus maximus (95, 100), with the greatest muscle activity occurring at 140% of shoulder width (91). However, maximal hip extension torque in these exercises occurs when the hip is flexed; torque progressively decreases during extension and is minimal at the finish of the movement. This is counter to maximal activation of the gluteus maximus, which occurs at the end range of hip extension (149). Indeed, EMG data show that the hip thrust produces significantly greater activation of the gluteus maximus compared to the squat (34). Moreover, gluteus maximus activity is diminished during combined hip and knee extension, although activation of the three *vasti* muscles (vastus lateralis, vastus intermedius, and vastus medialis) of the quadriceps is enhanced (152). Therefore, multi-joint lower-body movements might be best for inducing muscle damage in the gluteus maximus because peak activation occurs in the lengthened position, whereas an exercise such as the hip thrust is best for optimizing mechanical tension. Indeed, research shows that the gluteus maximus is optimally developed when performing deep versus shallow squats, corresponding to the lengthened position in which greater muscle damage occurs (75).

Single-joint hip extension exercises should also be incorporated for maximal development of the gluteus maximus. It is best to include a combination of all three lengths of accentuated force movements to cover the spectrum of mechanisms governing hypertrophy (33), as well as to target both upper and lower subdivisions of the musculature (123).

The primary action of the gluteus medius and gluteus minimus is to abduct the thigh. Frontal plane abduction movements, such as the cable hip side raise, are therefore needed

to target these muscles. The gluteus medius and minimus muscles also benefit from active external rotation during movement (26).

Anterior Thigh

The quadriceps are primary knee extensors and thus benefit from both multi-joint and single-joint lower-body movements. Multi-joint lower-body movements (e.g., the squat) have been found to elicit greater activation in the vasti muscles, whereas the knee extension targets the rectus femoris (42, 45). These results are consistent with research showing that multi-joint lower-body exercise maximally activates the quadriceps during deep knee flexion, whereas activation in open-chain knee extension is greatest during full extension (145). Additionally, as opposed to longitudinal studies employing isolated leg extension training (43), squat-only training has failed to display significant increases in rectus femoris muscle hypertrophy (75). Combined, the findings suggest a synergy between movements, which warrants combining exercises to achieve peak activation at varying muscle lengths.

Differences in muscular activation between lower-body multi-joint exercise may have hypertrophic implications. For example, the back squat and leg press show differential activation of the individual quadriceps heads (5). Similar findings have been demonstrated in variations of the squat, with the front squat showing greater activation of the vastus medialis than the back squat (153). Although heightened muscle activation does not necessarily translate into greater muscle growth, rotating lower-body multi-joint exercises over the course of a training cycle does seem to promote more symmetrical quadriceps development compared to performing the same movement on a volume-equated basis (48).

Stance width during multi-joint lower-body exercise does not appear to affect muscular activity in the quadriceps (91), nor does altering foot position (i.e., tibial rotation) from 30° inward rotation to 80° outward rotation (64, 95). On the other hand, there is evidence that foot position influences quadriceps activity in open-chain single-joint exercise, and that an

externally rotated position elicits greater activation of the rectus femoris (125). However, given that extreme rotation of the tibia can change normal patella tracking and potentially cause undesirable varus or valgus moments, the practical value of altering foot positions in an attempt to target aspects of the quadriceps remains questionable. Although some evidence indicates that a wider stance, particularly sumo style, can elicit greater adductor activation (91, 134), these findings are not universal (100).

Posterior Thigh

The hamstrings are a biarticular muscle complex. The semimembranosus, semitendinosus, and long head of the biceps femoris carry out both hip extension and knee flexion; the short head of the biceps femoris crosses only the knee joint and thus is purely a knee flexor. Contrary to popular belief, the hamstrings are only moderately active during multi-joint lower-body exercise, producing approximately half the amount of EMG activity as single-joint exercise (145, 150). This is consistent with the fact that when the hamstrings are shortening at the hip, they are lengthening at the knee, and vice versa. Their length thus remains fairly constant throughout performance, thereby limiting force output. In line with these findings, hypertrophy of the hamstrings is minimal following regular squat exercise (17, 75, 142), reinforcing the importance of the length–tension relationship in their development.

Single-joint exercises are required to fully stimulate the hamstrings. Exercises that involve hip extension (e.g., stiff-leg deadlift, good morning) and those that involve knee flexion (e.g., lying leg curl) are viable choices. Zebis and colleagues (156) found that the Romanian deadlift (a hip extension movement) targets the semitendinosus, whereas the lying leg curl (a knee flexion exercise) targets the biceps femoris. Moreover, there is evidence that knee flexion exercise produces greater activation of the lower aspect of the hamstrings (118), consistent with research showing that functional differences exist between proximal and distal compartments (141). Thus, both types of movements should be included for optimal muscular devel-

opment. The individual hamstring muscles can be further targeted by altering foot position during both hip extension (closed-chain) and knee flexion (open-chain) exercise. Internally rotating the foot targets the semitendinosus and semimembranosus, and external rotation favors the biceps femoris (83).

Lower Leg

The gastrocnemius and soleus (collectively known as the *triceps surae*) are the primary plantar flexors of the ankle joint and comprise the bulk of the muscular mass in the calf region. The gastrocnemius is a biarticular muscle that originates at the distal femur and fuses with the Achilles tendon to insert at the calcaneus. At the ankle, the gastrocnemius acts as a plantar flexor, whereas at the knee, it assists the hamstrings in flexion. Thus, straight-leg (knee) plantar flexion exercises (e.g., standing calf raise) place the gastrocnemius under maximal stretch and maximize force output (62). Alternatively, bent-leg (knee) plantar flexion exercises (e.g., seated calf raise) render the gastrocnemius actively insufficient and allow the uniarticular soleus to take over a majority of the work (62). There also is evidence that foot position can influence calf muscle activation: Turning the feet inward targets the lateral head of the gastrocnemius, whereas turning the feet outward targets the medial head (28, 88, 107), although the overall effect of this strategy on muscular activity is relatively modest and of questionable practical meaningfulness from a hypertrophy standpoint.

Abdominals

The rectus abdominis is the primary muscle responsible for carrying out spinal flexion. It spans from just below the sternum to the crest of the pubis. Instead of having a single muscular sheath, the rectus abdominis is partitioned by tendinous intersections. These fibrous bands of connective tissue compartmentalize the muscle into distinct segments that, when well-developed, give the abdominals the so-called “six pack” appearance.

Given its role in spinal flexion, variations of the crunch are viable options for dynamically

working the rectus abdominis. Although somewhat speculative, there is a sound rationale for performing traditional crunch variations to target the upper abdominal region and performing reverse crunch variations to develop the lower aspect of the muscle. This hypothesis is consistent with the anatomical design of the rectus abdominis. Not only do the tendinous intersections suggest some degree of functional independence of the muscle, but its upper and lower aspects are segmentally innervated by the ventral rami of the lower six or seven thoracic nerves (56), providing a further mechanism for selective activation. Indeed, professional tennis players demonstrate greater hypertrophy in the nondominant compared to the dominant side of the rectus abdominis, particularly in the more distal regions, indicating that humans can differentially recruit both sides of the rectus abdominis as well as the upper and lower regions of each muscle during exercise performance (110).

Electromyographic research investigating the ability for reverse crunch variations to enhance muscle activation in the lower abdominal region has produced conflicting results; some studies observe a beneficial effect (41, 111, 146) and others fail to note significant differences in activation between regions (30, 46, 78). A potential explanation for discrepancies between findings is that benefits may depend on consciously tilting the pelvis backward by drawing it up toward the umbilicus (posterior pelvic tilt) while performing the exercise. This was elegantly demonstrated by Sarti and colleagues (111), who found that activation of the lower abdominals was predicated on the participants' ability to initiate proper performance of a posterior pelvic tilt during the reverse crunch. Although some practitioners have cautioned that performing spinal flexion exercise is injurious to the discs (92), the body of evidence does not support such claims in people free of spine-related conditions (32).

The internal and external obliques assist the rectus abdominis in spinal flexion. However, they also are the primary muscles responsible for both rotation and lateral flexion of the spine. Thus, including exercises such as varia-

tions of side bends and rotational movements may help to optimize their development.

Isometric exercises also can develop the abdominal region. Planks and bridging movements statically work the musculature in a manner that can provide an additive abdominal stimulus. However, these movements are traditionally performed with body weight, and thus can be self-limiting based on a person's individual abilities; it can be difficult for well-trained people to overload the abdominal muscles using these exercises. To reap benefits, it is necessary to make the movements progressively more challenging by modifying aspects of performance. For example, the plank can be modified by moving the elbows superiorly toward the ears and engaging a posterior pelvic tilt; this significantly increases activation in the rectus abdominis and oblique muscles (117).

Periodization

Hypertrophy-oriented resistance training program design is thought to benefit from a periodized approach (63). Simply stated, the goal of periodization is to optimize a given fitness component over time. This is accomplished by manipulating program variables to create consistent improvement in the target outcome while minimizing the potential for plateau or regression.

Periodization is loosely based on Selye's general adaptation syndrome (GAS) theory (36), which proposes that the body undergoes a three-stage reaction to stress: alarm, resistance, and exhaustion (figure 8.6) (124). An applied example of the GAS theory is the body's response to a virus. Initially, exposure to the virus causes an alarm reaction in which the immune system is mobilized to counteract the stressor. If the immune defense is sufficiently strong, the virus is quelled and the body becomes resistant to subsequent exposure. However, if the virus overwhelms the immune response, health continues to decline and leads to severe illness or even death.

Given that intense physical activity is a potent stressor, the GAS theory has relevance to exercise. Performance of rigorous resistance

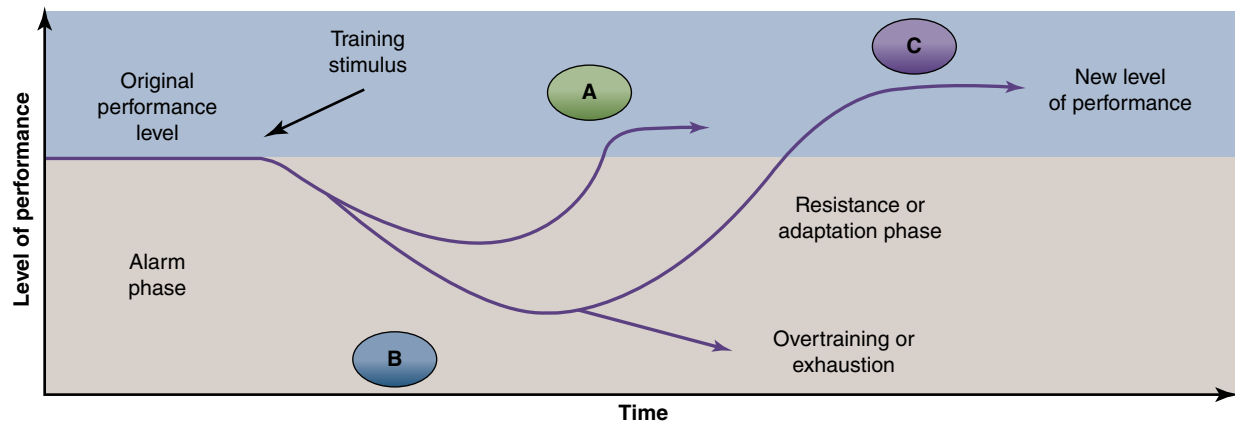


FIGURE 8.6 Illustration of Selye's general adaptation syndrome theory. A = typical training; B = overtraining; C = overreaching or supercompensation.

Adapted by permission from A.C. Fry, "The Role of Training Intensity in Resistance Exercise Overtraining and Overreaching," in *Overtraining in Sport*, edited by R.B. Kreider, A.C. Fry, and M.L. O'Toole (Champaign, IL: Human Kinetics, 1998), 114.

training initiates an alarm response in the body that ultimately leads to increases in protein synthesis and other anabolic processes. Under ideal circumstances, the exercise stress is sufficient to cause a supercompensatory response that results in greater muscle protein accretion. If the applied stress does not progressively challenge the neuromuscular system sufficiently, a plateau ensues and no further increases in growth occur. Alternatively, if the stress is repeatedly too great for the body's recovery processes, the response is maladaptive and leads to an overtrained state. While there is a large interindividual variation in the stressor response, emerging evidence indicates that high levels of stress applied persistently over time downregulates the immune system, motor coordination, cognition, mood, metabolism, and hormonal function (69), which in turn has detrimental effects on muscular adaptations. To avoid the negative consequences of nonfunctional overreaching/overtraining and ensure ongoing increases in growth, lifters can benefit from periodizing their exercise programs over time (8, 155).

Periodization Models

An array of periodization models have been proposed to maximize muscular adaptations to resistance training. Of these models, three

have been studied with respect to their effects on muscle hypertrophy: traditional linear periodization, nonlinear (undulating) periodization, and reverse periodization. This section provides an overview of the research on each of these models.

It should be noted that periodization is a concept, not a defined system of training. Thus, there are virtually unlimited ways to structure a periodized program based on a person's unique needs and abilities. Given that all training variables can be manipulated, and given the plethora of possible combinations of manipulation, the ability to draw practical inferences from research is limited. So although a logical rationale exists for the use of periodization as a strategy to help maximize hypertrophy, multiple approaches remain viable options.

Traditional Linear Periodization

The origins of periodization can be traced back to the 1950s. Matveyev is widely credited with developing the *traditional linear periodization model* to prepare athletes for Olympic competition (131). The linear model is made up of three basic phases: the *macrocycle*, which encompasses an entire training period generally ranging from 6 months to several years; the *mesocycle*, which splits the macrocycle into at least two subdivisions lasting from several weeks

PRACTICAL APPLICATIONS**IS THERE A BEST TIME OF DAY TO WORK OUT?**

It is well established that biorhythms can influence the performance of daily tasks. This holds true for most performance-oriented qualities. From a muscular strength standpoint, the time of day at which performance peaks (i.e., the acrophase) appears to occur in the evening hours, somewhere around 6 p.m. (58). It therefore has been proposed that resistance training should be carried out later in the day to take advantage of this phenomenon. Conceivably, higher strength levels should enhance mechanical tension during training, translating into greater muscular gains.

Some acute data support the concept of training based on the purported strength acrophase. For example, Burley and colleagues (23) observed a superior anabolic response to evening resistance exercise compared to the same workout performed in the morning. However, other research refutes such findings, showing similar increases in p70^{S6K} phosphorylation following resistance training bouts performed in the morning versus evening hours (122). Importantly, these studies only looked at the response to a single bout of exercise, and thus do not take into account how adaptations might be affected longitudinally over time.

A meta-analysis by Grgic and colleagues (58) sought to determine whether time of day affected long-term, exercise-induced muscle hypertrophy. Consistent with commonly held beliefs, results indicated that individuals tend to display greater baseline levels of strength in the evening hours compared to the morning. However, analysis of findings showed that when training is consistently performed in the morning, these differences even out so that strength becomes similar to that during evening training. In other words, people adapt to the time of day at which they train, and hence see a change in their acrophase. This finding seemingly indicates that time of day is an irrelevant consideration from a performance standpoint; over time, mechanical tension should not be affected by whether you train in the morning or evening.

Consistent with strength outcomes, analysis of hypertrophic changes shows similar training-induced increases in muscle size irrespective of whether training is carried out early or later in the day. It should be acknowledged that only 5 of the 11 studies meeting inclusion criteria for the meta-analysis assessed hypertrophy. Thus, caution must be used when interpreting these findings because current research is insufficient to draw strong conclusions on the topic.

Considering the totality of current evidence, it is misguided to blindly train based on the concept of an acrophase. Rather, personal preference and convenience should dictate when a person chooses to exercise. As a general rule, those who initially do not respond well to training at a given time of day will adapt and become similarly proficient with consistent adherence to that schedule. That said, it is possible, if not likely, that some individuals may not adapt well to a change in workout schedule, regardless of how long training is carried out at the alternative time of day. Thus, people should be cognizant of their performance and make adjustments accordingly. Time of day would be one factor to consider if performance regresses over time.

to months; and the *microcycle*, which further subdivides the mesocycle into weekly phases focused on daily training variations. In the classic linear model, intensity and volume are inversely structured so that mesocycles progress from periods of high volume and low intensity to periods of low volume and high intensity. A typical three-phase linear mesocycle begins with a hypertrophy or muscle endurance phase, or both, in which intensities of load are 60% to 75% of 1RM (10 to 20 repetitions). Next is a strength phase in which loading intensities range from 80% to 90% of 1RM (4 to 8 repetitions). The final mesocycle focuses on strength and power by increasing intensities even further, approaching or exceeding 95% of 1RM (2 to 5 repetitions). Each increase in intensity is met with a corresponding reduction in training volume to accommodate the greater stress on the neuromuscular system. Ultimately, the person peaks at the end of the final mesocycle so that the training outcomes transfer to competition.

Several studies have been carried out to determine whether periodizing a resistance training program enhances muscle growth, and results have been mixed. A recent systematic review of the topic identified 12 studies that compared hypertrophic changes in periodized versus nonperiodized resistance training programs. After taking into account the body of literature, no clear benefit was seen for periodizing training as a strategy to elicit gains in muscle mass. When attempting to draw evidence-based conclusions on periodization, however, it is important to note several important limitations of current research on the topic.

For one, the vast majority of periodization research has been carried out on untrained individuals, with only two of the included studies involving subjects with previous resistance training experience. This is problematic because the adaptations in the initial phase of training are primarily directed toward neural improvements in recruitment, rate coding, and synchronization within and between muscles. If anything, naive trainees would benefit from consistently performing the same routine for the first month or two in order to better ingrain motor patterns for exercise performance; only

after an individual gains proficiency in lifting technique would there be a potential benefit to systematically manipulating variables. Consistent with this point, De Souza and colleagues (39) found that untrained subjects similarly increased quadriceps cross-sectional area over the first 6 weeks of engaging in either a periodized or nonperiodized routine; however, after training an additional 6 weeks in their respective programs, only the group performing the periodized routine continued to realize hypertrophic gains.

Equally important, the length of most periodization studies is relatively short, generally lasting a maximum of 12 weeks. Given that overtraining tends to manifest over longer periods, the majority of studies simply aren't properly designed to investigate the impact of periodization on hypertrophic outcomes. In the longest periodization study to date, Kraemer and colleagues (73) reported that female tennis players gained a significantly greater amount of fat-free mass following a periodized compared to a nonperiodized resistance training program over 9 months (3.3 vs. 1.6 kg, or 7.2 vs. 3.5 lb, respectively). Results must be viewed with circumspection, however, as the skinfold method was employed to estimate fat-free mass.

Another issue in current research on the topic is the predominant use of indirect measures to assess hypertrophy; only 3 of the 12 studies meeting inclusion criteria employed a site-specific measurement technique. As noted in chapter 3, site-specific modes display a greater ability to detect the rather subtle changes that occur over relatively short-term training studies. In the first study on the topic to assess muscle growth in resistance-trained individuals using a site-specific measure (ultrasound), my lab (114) randomized subjects to an 8-week resistance training protocol performing sets at either 8RM to 12RM for all sessions or undulating loading into heavy (3RM to 5RM), moderate (8RM to 12RM), and light (20RM to 30RM) sessions carried out on alternating days over the course of each week. Although no statistical differences were noted in muscular outcomes, the group that undulated its training program showed a modestly greater magnitude of increase in muscle thickness for the biceps and

triceps from pre- to post-study. The practical meaningfulness of these variances remains questionable, but the short study duration raises the possibility that greater hypertrophic gains may be realized over time by periodizing variables in this manner.

Thus, although the research on the topic remains equivocal and is confounded by the aforementioned limitations, the literature does seem to suggest a potential benefit for systematically manipulating variables over time to maximize hypertrophic adaptations, and the logical basis of the approach advocates employing periodization for the goal of muscle building. Moreover, considerable evidence shows that periodization elicits greater gains in strength than nonperiodized approaches do (1, 94, 96, 132, 147). Given that mechanical tension is a primary driving force for muscle protein accretion (116), a case can be made that greater increases in strength alone would facilitate superior hypertrophic gains over time. Table 8.1 provides a summary of the research related to periodized versus nonperiodized programs.

Nonlinear (Undulating) Periodization

Several variations to the original periodization model have been proposed to enhance results. One of the most popular is the concept of *nonlinear periodization*, often referred to as *undulating periodization*, which was first introduced into the literature by Poliquin (103). Nonlinear periodization is thought to address inherent issues with the traditional model—namely, that progressive increases in load intensity do not allow sufficient time for regeneration, thus placing undue stress on the body over extended periods and increasing the potential for overtraining (103). Moreover, the hypertrophic gains obtained during the early phases of training are not well maintained because volume—a primary driver of hypertrophy—is progressively decreased over the latter phases of the linear macrocycle. To account for these drawbacks, nonlinear periodized programs vary volume and intensity in an undulatory manner. The phases are therefore much shorter in the nonlinear approach. Poliquin (103) originally proposed alternating phases of accumulation

and intensification biweekly to optimize a given fitness outcome without overtaxing bodily systems. A popular modification to this approach is the *daily undulating periodization* (DUP) model. Typically, DUP involves alternating heavy-, moderate-, and light-load sessions over the course of a week.

Several studies have been carried out to directly compare the hypertrophic adaptations of volume-equated linear and nonlinear periodization models (9, 38, 61, 72, 94, 105, 126, 129); see table 8.2 on page 197 for a summary. Of these studies, only one reported significant differences in the models; the nonlinear approach produced superior increases in the thickness of the elbow flexors and elbow extensors in untrained young men (126). In one of the more well-controlled studies on the topic, Pelzer and colleagues (102) found that linear and nonlinear periodized routines that were equated for total training volume load, number of repetitions within each loading zone, range of motion, and time under tension produced similar increases in quadriceps growth. Unsurprisingly, therefore, meta-analytic data indicate that hypertrophy is similar between the two approaches (57). Taking the body of literature as a whole, both linear and nonlinear models seem to be equally viable options for promoting increases in muscle growth.

Reverse Periodization

Another variation of the traditional periodization model specifically designed to maximize hypertrophy is *reverse periodization*. As previously mentioned, the traditional linear model involves progressive reductions in training volume to account for corresponding increases in load. Considering the strong dose–response relationship between volume and hypertrophy, this seemingly is counterproductive for maximizing muscle mass in the peak phase of the macrocycle. Reverse periodization addresses this issue by placing a hypertrophy mesocycle at the end of the macrocycle so that volume is relatively high at the point at which a peak is desired.

Research comparing the hypertrophic adaptations of linear and reverse linear models is sparse (see table 8.3 on page 200). In one of

TABLE 8.1 Summary of Hypertrophy Training Studies Investigating Periodized Versus Nonperiodized Programs

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Ahmadizad et al. (1)	32 trained young men	Random assignment either to a nonperiodized, linear periodized, or undulating periodized total-body resistance training program. Training was carried out 3 times per week.	8 weeks	BIA	No significant differences in lean mass between conditions
Baker et al. (9)	22 resistance-trained young men	Random assignment to a linear periodized, undulating periodized, or nonperiodized split-body resistance training protocol. The linear protocol progressively increased load from 10RM to 3RM; the undulating protocol rotated every other week between 3RM and 10RM; the nonperiodized protocol performed 6RM every session. All subjects performed multiple sets of multiple exercises each session. Training was carried out 3 days per week.	12 weeks	Skinfold technique	No significant differences in lean mass between conditions
Conlon et al. (31)	33 untrained older men and women	Random assignment to either a nonperiodized, block periodized, or daily undulating periodized resistance training program. The nonperiodized group performed repetitions at 10RM, and the periodized groups varied loading between 5RM and 15RM. All groups performed 3 sets of 6 different exercises carried out 3 days per week.	22 weeks	DXA	No significant differences in FFM between conditions
De Souza et al. (39)	25 untrained young men	Random assignment to either a nonperiodized, block periodized, or daily undulating periodized lower-body resistance training program. The nonperiodized group performed 2 to 3 sets of 8RM, and the periodized groups performed 2 to 4 sets of 4RM to 12RM. All groups trained 2 days per week.	12 weeks	MRI (quadriceps)	All groups increased quadriceps CSA over the first 6 weeks of the study; however, only the periodized groups increased quadriceps hypertrophy over the final 6 weeks of the study.
Fink et al. (47)	21 untrained young men	Random assignment to either a high-load nonperiodized, low-load nonperiodized, or a weekly undulating periodized resistance training program; the periodized group rotated high- and low-load training every 2 weeks. All groups performed 3 sets of biceps curls carried out 3 times per week.	8 weeks	MRI (biceps brachii)	No significant differences in elbow flexor CSA between groups
Hunter et al. (65)	28 untrained older men	Random assignment to either a high-load nonperiodized or daily undulating periodized total-body resistance training program. The nonperiodized group trained at 80% of 1RM, and the periodized group trained at 80%, 65%, and 50% of 1RM. Training consisted of 2 sets per exercise carried out 3 times per week.	25 weeks	BodPod	No significant differences in lean mass between conditions

(continued)

Table 8.1 (continued)

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Kramer et al. (74)	39 young men	Random assignment to either a nonperiodized single-set to failure, nonperiodized multiset, or weekly undulating resistance training program. The nonperiodized group performed repetitions at 10RM, and the periodized group varied loading and volume over the course of the study period. All groups trained 3 days per week.	14 weeks	Skinfold technique	No significant differences in lean mass between conditions
Kraemer et al. (73)	19 untrained young women	Random assignment to either a nonperiodized or daily undulating periodized total-body resistance training program. The nonperiodized group trained with 8RM to 10RM, and the periodized group varied loading from 4RM to 15RM. Both groups performed 3 sets per exercise carried out 3 days per week.	36 weeks	Skinfold technique	No significant differences in lean mass between conditions. Relative increases in lean mass favored the periodized condition.
Marx et al. (89)	22 untrained young women	Random assignment to either a nonperiodized or a daily undulating resistance training program. The nonperiodized group performed a single set of 8 to 12 repetitions to muscular failure 3 days per week in circuit fashion; the periodized group performed 2 to 4 sets of 3 to 15 repetitions carried out 4 days per week.	24 weeks	Hydrostatic weighing	Greater increases in FFM for the periodized condition
Monteiro et al. (94)	27 resistance-trained young men	Random assignment to a linear periodized, undulating periodized, or nonperiodized split-body resistance training protocol. Those in the linear protocol progressively increased the load from 12RM to 15RM up to 4RM to 5RM; those in the undulating protocol rotated between 4RM and 15RM over each microcycle, and those in the nonperiodized protocol performed 8RM to 10RM every session. Multiple sets were performed for 13 exercises. Training was carried out 4 days per week.	12 weeks	Skinfold technique	No significant differences in lean mass between conditions, although the nonperiodized group lost lean mass while the periodized groups showed slight gains.
Schiotz et al. (112)	14 untrained young men	Random assignment to either a nonperiodized or a weekly undulating total-body resistance training program. The nonperiodized group trained at 80% of 1RM, and the periodized group trained with loads ranging from 50% to 105% of 1RM. Both groups trained 4 days per week.	10 weeks	Skinfold technique	No significant differences in lean mass between conditions
Schoenfeld et al. (114)	19 trained young men	Random assignment to either a nonperiodized or a daily undulating total-body resistance training program. The nonperiodized group trained with 8RM to 12RM, and the periodized group trained across a spectrum of loading zones (3RM to 30RM). Both groups performed 3 sets of each exercise carried out 3 days per week.	8 weeks	Ultrasound (elbow flexors, elbow extensors, quadriceps)	No significant differences in muscle thickness between conditions. The magnitude of increase in upper-extremity muscle thickness modestly favored the periodized group.

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Stone et al. (133)	20 young men (training status not disclosed)	Random assignment to either a periodized or a nonperiodized split-body resistance training protocol. The periodized group trained with 5 sets of 10 reps in weeks 1 through 3, 5 sets of 5 reps in week 4, 3 sets of 3 reps in week 5, and 3 sets of 2 reps in week 6; the nonperiodized group performed 3 sets of 6 reps each session. All subjects performed 6 multi-joint exercises over the course of 3 weekly training sessions.	6 weeks	Underwater weighing	Significantly greater increases in lean body mass for the periodized condition
Souza et al. (129)	31 recreationally active young men	Random assignment to a linear periodized, undulating periodized, or nonperiodized lower-body resistance training protocol. Those in the linear protocol performed 12RM in weeks 1 through 4 and 8RM in weeks 5 and 6; those in the undulating protocol rotated between 12RM and 8RM in weeks 1 through 4 and then 6RM to 10RM in weeks 5 and 6; those in the nonperiodized protocol performed 8RM every session. Multiple sets were performed for two exercises. Training was carried out 2 days per week.	6 weeks	MRI	No significant differences in quadriceps CSA between conditions

Abbreviations: RM = repetition maximum; MRI = magnetic resonance imaging; CSA = cross-sectional area; BIA = bioelectrical impedance analysis; DXA = dual X-ray absorptiometry; FFM = fat-free mass.

TABLE 8.2 Summary of Hypertrophy Training Studies Investigating Linear Versus Nonlinear Periodization

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Ahmadizad et al. (1)	32 trained young men	Random assignment either to a nonperiodized, linear periodized, or undulating periodized total-body resistance training program. Training was carried out 3 times per week.	8 weeks	BIA	No significant differences in lean mass between conditions
Baker et al. (9)	22 trained young men	Random assignment to a linear periodized, undulating periodized, or nonperiodized total-body resistance training protocol. Those in the linear protocol progressively increased load from 10RM to 3RM; those in the undulating protocol rotated every other week between 3RM and 10RM; those in the nonperiodized protocol performed 6RM every session. All subjects performed multiple sets of multiple exercises each session. Training was carried out 3 days per week.	12 weeks	Skinfold technique	No significant differences in lean mass between conditions
Buford et al. (22)	28 untrained young men and women	Random assignment to either a linear periodized, daily undulating, or weekly undulating total-body resistance training program. Training was carried out 3 times per week.	9 weeks	Circumference and skinfold technique	No significant differences in changes in muscle girth between groups

(continued)

Table 8.2 (continued)

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
de Lima et al. (38)	28 untrained young women	Random assignment to either a linear or undulating periodized resistance training program. The linear protocol increased load each week for 4 weeks from 30RM to 25RM up to 20RM to 15RM and then repeated this sequence for the balance of the study; the undulating protocol alternated weekly between 25-30RM and 15-20RM. Multiple sets were performed for 16 exercises in split-body fashion. Training was carried out 4 days per week.	12 weeks	Skinfold technique	No significant differences in lean mass between conditions
Harries et al. (61)	26 recreationally trained adolescent males	Quasi-experimental random assignment to either a linear or undulating periodized resistance training program. The linear protocol progressively increased load each week; the undulating protocol varied between a higher- and lower-repetition day each week. Multiple exercises were performed, but only the squat and bench press were periodized. All training was carried out twice per week.	12 weeks	BIA	No significant differences in skeletal muscle mass between conditions
Kok et al. (72)	20 untrained young women	Random assignment to either a linear or undulating periodized resistance training program. The linear protocol progressively increased load every 3 weeks from 10RM to 6RM to 3RM; the undulating protocol varied loading each week from 10RM to 6RM to 3RM and then repeated this cycle over the course of the study. Three sets were performed for 10 exercises carried out 3 days per week.	9 weeks	Ultrasound (quadriceps)	No significant differences in quadriceps femoris thickness between conditions
Monteiro et al. (94)	27 trained young men	Random assignment to a linear periodized, undulating periodized, or nonperiodized split-body resistance training protocol. Those in the linear protocol progressively increased load from 12RM to 15RM up to 4RM to 5RM; those in the undulating protocol rotated between 4RM and 15RM over each micro-cycle; those in the nonperiodized protocol performed 8RM to 10RM every session. Multiple sets were performed for 13 exercises. Training was carried out 4 days per week.	12 weeks	Skinfold technique	No significant differences in lean mass between conditions, although the nonperiodized group lost lean mass, while the periodized groups showed slight gains.
Prestes et al. (105)	40 trained young men	Random assignment to either a linear or undulating periodized resistance training program. The linear protocol increased load each week for 4 weeks from 12RM to 10RM to 8RM to 6RM and then repeated this sequence for the balance of the study; the undulating protocol alternated weekly between 10RM to 12RM and 6RM to 8RM. All subjects performed 3 sets of multiple exercises carried out 4 days per week.	12 weeks	Skinfold technique	No significant differences in lean mass between conditions

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Simao et al. (126)	30 recreationally trained young men	Random assignment to either a linear or undulating periodized resistance training program. The linear protocol focused on local muscular endurance the first 4 weeks (2 × 12RM), hypertrophy the next 4 weeks (3 × 8RM), and strength the final 4 weeks (4 × 3RM); the undulating protocol varied these components every 2 weeks for 6 weeks and then repeated this schedule the next 6 weeks. All subjects performed multiple sets of 4 upper-body exercises.	12 weeks	Ultrasound (elbow flexors, elbow extensors)	No significant differences in thickness of the biceps or triceps were noted between conditions, but only the undulating group showed significant increases from baseline in these measures.
Souza et al. (129)	31 recreationally active young men	Random assignment to a linear periodized, undulating periodized, or nonperiodized lower-body resistance training protocol. Those in the linear protocol performed 12RM in weeks 1 through 4 and 8RM in weeks 5 and 6; those in the undulating protocol rotated between 12 and 8 reps in weeks 1 through 4 and then between 6 and 10 reps in weeks 5 and 6; those in the nonperiodized protocol performed 8RM every session. Multiple sets were performed for 2 exercises. Training was carried out 2 days per week.	6 weeks	MRI (quadriceps)	No significant differences in quadriceps CSA between conditions

Abbreviations: BIA = bioelectrical impedance analysis; RM = repetition maximum; CSA = cross-sectional area; MRI = magnetic resonance imaging.

the few controlled studies on the topic, Prestes and colleagues (104) randomized a group of young women experienced in resistance training to perform either a traditional periodized program in which loads were progressively increased from 12RM to 14RM up to 4RM to 6RM or a program in which the progression was reversed (from 4RM to 6RM down to 12RM to 14RM). Both groups performed 3 sets of multiple exercises for the whole body, and training occurred 3 days per week over 12 weeks. Body composition, as assessed by the skinfold method, showed that subjects in the linear periodized group significantly increased fat-free mass by approximately 7%, whereas those in the reverse linear periodized group had nonsignificant increases of approximately 4%. Although these results are intriguing and somewhat counterintuitive, the use of skinfold measurement limits the ability to draw definitive conclusions about the difference in hypertrophic effects of the two periodization models.

KEY POINT

Both linear and nonlinear models of periodization seem to be equally viable for maximizing hypertrophy. Despite a logical basis, reverse periodization has not been shown to be more effective, but more research is needed to be able to draw definitive conclusions.

Deloading Periods

The accretion of muscle proteins requires that the body be repeatedly challenged beyond its present state over time. However, persistently overtaxing the body's resources with excessive training and insufficient recovery ultimately leads to an *overtrained state* (i.e., the exhaustion phase of GAS). The upshot is an increase in the expression of catabolic proteins (atrogin-1) and a reduction in anabolic factors (MyoD,

TABLE 8.3 Summary of Hypertrophy Training Studies Investigating Linear Versus Reverse Linear Periodization

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Prestes et al. (104)	20 trained young women	Random assignment to either a linear periodization protocol beginning with 12RM to 14RM and progressively increasing loads to finish with 4RM to 6RM or a reverse linear protocol beginning with 4RM to 6RM and progressively decreasing loads to finish with 12RM to 14RM. All subjects performed 3 sets of 8 or 9 exercises 3 days per week.	12 weeks	Skinfold measurements	Greater increases in FFM with linear periodization

Abbreviations: RM = repetition maximum; FFM = fat-free mass.

myogenin, and IGF-1), and a corresponding decrease in muscle cross-sectional area (3). There is evidence that such negative complications can be avoided by taking short breaks from training. Animal research shows that chronic resistance training suppresses the phosphorylation of intracellular anabolic signaling, but signaling is restored after a brief period of detraining (98). Ogasawara and colleagues (97) demonstrated that taking a 3-week break from training at the midpoint of a 15-week resistance training program did not interfere with muscular adaptations. Follow-up work from the same lab found that repeated 3-week detraining and 6-week retraining cycles produced improvements in muscle cross-sectional area that were similar to those resulting from continuous resistance training over a 6-month period (99).

Rather than taking time off from training, people may be able to enhance muscular adaptations via a *deloading period*—that is, systematically reducing training intensity or volume, or both. When properly executed, deloading promotes restoration and rejuvenation in a manner that facilitates continued progress (18). Unfortunately, no studies to date have attempted to quantify the extent of reductions in either volume or intensity (or both) to best promote hypertrophic gains. A 3:1 ratio (in weeks) of training and deloading is generally a good starting point; modifica-

tions should then be made depending on the needs and abilities of the individual.

Periodizing Intensity of Load

As previously explained, sessions can be partitioned into loading zones encompassing heavy loads (1RM to 5RM), moderate loads (8RM to 12RM), and light loads (≥ 20 RM). A periodized approach to this variable can be carried out using either a linear or undulating model. Note that loading can also be varied within a given session. For example, a lower-body routine could include the squat performed at 5RM, leg press performed at 10RM, and leg extension at 15RM. Alternatively, pyramid systems can use different loading zones for the same exercise over the course of a fixed number of sets. Both ascending (loads progressively increase with each subsequent set) and descending (loads progressively decrease with each subsequent set) pyramids are viable options. Research indicates that a pyramid performed with a wide loading zone (sets of 15RM, 10RM, and 5RM) results in greater increases in skeletal muscle mass compared to a narrower loading zone (sets of 12RM, 10RM, and 8RM) (40).

Table 8.4 on page 203 illustrates a strategy for varying loads across a 3-day-per-week undulating program in which all muscles are trained in a session. Table 8.5 on page 204 expands the undulating program to a 4-day upper/lower split. Note that in this scenario all loading ranges are trained over the course

PRACTICAL APPLICATIONS**DOES SLEEP AFFECT MUSCLE GROWTH?**

Sleep is a basic human need. It is considered important to human physiology and cognition, and thus has relevance to recovery from exercise. Three primary factors determine the recuperative outcome of sleep: the duration (total sleep time), quality, and phase (circadian timing) of sleep (109). Research indicates that disruptions in sleep-related factors can have a negative impact on exercise performance (50). However, the impact of sleep on muscle hypertrophy has received less attention.

A recent review of literature concluded that shiftwork, which tends to disrupt sleep patterns, is associated with a negative influence on skeletal muscle health, including a decrease in muscle protein synthesis and elevations in proteolysis (2). However, these workers also experience alterations in other lifestyle factors such as exposure to natural and artificial light, as well as nutritional intake. Decreased sleep quality also is a symptom of overtraining syndrome, leading to speculation that dysregulated sleep patterns may be a trigger for overtraining (25). But again, multiple other factors are purportedly involved in the condition as well (e.g., increased duration of work or study; decreased calorie, carbohydrate, and protein intakes; worsened mood states; and decreased hydration), thereby confounding the ability to draw strong conclusions about the specific role of sleep.

Several observational studies have shown an association between poor sleep and low levels of lean mass (21, 71, 135), although these findings are not universal (101). Other observational research indicates that effects of short sleep duration may be sex specific, with greater detriments on lean mass in women compared to men. Moreover, research shows that a year of melatonin treatment positively affected lean mass gains, presumably by helping to regulate sleep patterns (4). Interestingly, evidence also shows that excessive sleep duration correlates with lower lean mass (70, 135), suggesting there may be a sweet spot for optimizing muscle growth. However, while these studies provide intriguing evidence on the topic, they are correlational in nature and thus cannot be used to draw causality.

Controlled research investigating the effects of sleep deprivation in combination with regular resistance training on muscle hypertrophy is scant. An 8-week rodent study (93) randomized rats to one of five conditions: (1) control, (2) sleep deprived, (3) resistance trained, (4) sham (trained on the apparatus without load), or (5) resistance trained and sleep deprived. Sleep deprivation consisted of placing animals on a platform in a stainless-steel reservoir filled with water up to 1 centimeter (0.4 inch) below the surface of the platforms; if an animal fell asleep, it would make contact with the water and thus be stirred to wake up. Sleep deprivation was carried out for 96 continuous hours. Resistance training involved loaded stair climbing with resistance progressively increased to continually challenge muscular abilities. Results showed that while resistance training attenuated atrophy in sleep-deprived rodents, muscle mass only returned to baseline values; the resistance-trained group with normal sleep realized substantially greater hypertrophic increases (~10%). Moreover, testosterone and IGF-1 levels were blunted, and cortisol levels were elevated in sleep-deprived animals compared to those with normal sleep. Although the findings are intriguing, it should be noted that the extent of sleep deprivation was severe (96 hours), and thus, findings cannot be generalized to predict how restricted sleep (disrupted sleep–wake cycle), as commonly seen in human populations, affects hypertrophic adaptations. To

(continued)

Does Sleep Affect Muscle Growth? *(continued)*

date, no studies have endeavored to directly examine the effects of restricted sleep on exercise-induced changes in muscle growth in humans; thus, it is not possible to draw strong inferences on the practical implications of the topic.

Overall, current evidence suggests that sleep plays a role in muscle development. However, sleep function is complex, involving both qualitative and quantitative aspects, and a variety of factors thus must be considered when attempting to draw inferences about sleep recommendations to optimize hypertrophy. For example, while a compelling body of evidence indicates that sleep deprivation is detrimental to exercise performance, the effects of restricted sleep patterns are less clear. Thus, potential detriments to hypertrophy resulting from performance impairment warrant consideration in this context. Moreover, despite guidelines suggesting that humans require 7 to 9 hours of sleep each night, anecdotal evidence suggests that people can thrive across a fairly wide range of durations, with some needing more sleep and others less. Ultimately, requirements are likely individual. The best recommendation therefore is to gauge sleep needs based on how a person feels during training; perceptions of fatigue and lethargy could indicate that sleep-related issues are interfering with results.

of 10 days as opposed to 1 week in the 3-day full-body program.

Table 8.6 on page 205 illustrates a modified linear approach to varied loading for hypertrophy. The length of each mesocycle is generally between 1 and 3 months, but it can be shorter or longer depending on the person's goals and abilities. Note that the hypertrophy mesocycle is at the end of the macrocycle so that growth peaks at this time.

Figure 8.7 on page 209 shows how a step-loading approach can be employed in the context of a linear model. Step loading involves a progressive increase in intensity of load over a period of weekly microcycles followed by a deloading period of substantially reduced intensity. This structure creates a wavelike loading pattern that allows the use of a broad spectrum of repetitions within a target repetition range while at least theoretically reducing the potential for overtraining. The example in figure 8.7 is specific to a hypertrophy mesocycle, but the concept is applicable to any loading zone.

Periodizing Volume and Frequency

A clear dose–response relationship has been found between volume and hypertrophy; higher training volumes correlate with greater muscle protein accretion, at least up to a given threshold. However, consistently training with high volumes will inevitably overtax recuperative abilities, leading to an overtrained state. Excessive volume has been shown to cause greater neuroendocrine disruptions than consistently training at very high intensities (49). A logical solution is to increase training volume progressively over the course of a training cycle. The cycle begins with a maintenance volume dose, and then volume is systematically increased to culminate in a brief cycle at the highest tolerable dose that elicits functional overreaching, thereby eliciting a supercompensatory hypertrophic response. After a period of active recovery, the process repeats, beginning with the maintenance dose

TABLE 8.4 Sample 3-Day Undulating Periodized Program

Exercise	Sets	Repetitions	Rest interval
Monday (heavy)			
Bench press	4 or 5	3 to 5	3 minutes
Bent barbell row	4 or 5	3 to 5	3 minutes
Military press	4 or 5	3 to 5	3 minutes
Squat	4 or 5	3 to 5	3 minutes
Romanian deadlift	4 or 5	3 to 5	3 minutes
Wednesday (moderate)			
Incline press	3 or 4	8 to 12	2 minutes
Lat pulldown	3 or 4	8 to 12	2 minutes
Upright row	3 or 4	8 to 12	2 minutes
EZ curl	2 or 3	8 to 12	2 minutes
Overhead triceps extension	2 or 3	8 to 12	2 minutes
Leg press	3 or 4	8 to 12	2 minutes
Seated leg curl	3 or 4	8 to 12	2 minutes
Standing calf raise	2 or 3	8 to 12	2 minutes
Kneeling abdominal cable crunch	2 or 3	8 to 12	2 minutes
Friday (light)			
Dumbbell incline fly	2 or 3	15 to 25	30 to 60 seconds
Seated cable row	2 or 3	15 to 25	30 to 60 seconds
Machine lateral raise	2 or 3	15 to 25	30 to 60 seconds
Dumbbell hammer curl	2 or 3	15 to 25	30 to 60 seconds
Cable pushdown	2 or 3	15 to 25	30 to 60 seconds
Knee extension	2 or 3	15 to 25	30 to 60 seconds
Hyperextension	2 or 3	15 to 25	30 to 60 seconds
Seated calf raise	2 or 3	15 to 25	30 to 60 seconds
Reverse crunch	2 or 3	15 to 25	30 to 60 seconds

Concepts adapted from B.J. Schoenfeld, 2013, *The M.A.X. Muscle Plan* (Champaign, IL: Human Kinetics, 2013).

to help reset the muscles' volume sensitivity. In support of this approach, research from Eduardo De Souza's lab shows that the individual response to resistance training volume may be predicated on the amount of volume the person had been performing previously (personal correspondence). Specifically, those who realized the greatest hypertrophic gains increased their training volume by an average of 6.6 sets compared to the lowest responders who increased volume by 1.8 sets. These results give credence to the possibility that a period of lower training volume can prime the muscles to respond better to future higher volume cycles.

Evidence that inducing a state of functional overreaching via a brief period of high-volume

training may enhance hypertrophy was recently provided by Bjornsen and colleagues (16), who subjected untrained men and women to two 5-day blocks of 7 blood flow restriction sessions (training was carried out either daily or twice daily), with the blocks separated by a 10-day recovery period. Training consisted of 4 sets of unilateral knee extensions to volitional failure at 20% of 1RM per session. Results showed a delayed hypertrophic response, whereby muscle size initially decreased after the first block and then rebounded to supercompensate, with increases in muscle size peaking 10 days after cessation of the second block.

The importance of limiting functional overreaching attempts to brief cycles was demonstrated

TABLE 8.5 Sample 4-Day Undulating Periodized Program

Exercise	Sets	Repetitions	Rest interval
Week 1			
<i>Monday (heavy lower)</i>			
Squat	5 or 6	3 to 5	3 minutes
Deadlift	5 or 6	3 to 5	3 minutes
Leg press	5 or 6	3 to 5	3 minutes
Glute to ham raise	5 or 6	3 to 5	3 minutes
<i>Tuesday (heavy upper)</i>			
Bench press	5 or 6	3 to 5	3 minutes
Weighted pull-up	5 or 6	3 to 5	3 minutes
Standing push-press	5 or 6	3 to 5	3 minutes
Barbell bent row	5 or 6	3 to 5	3 minutes
<i>Thursday (moderate lower)</i>			
Front squat	3 or 4	8 to 12	2 minutes
Bulgarian split squat	3 or 4	8 to 12	2 minutes
Barbell hip thrust	3 or 4	8 to 12	2 minutes
Romanian deadlift	3 or 4	8 to 12	2 minutes
Lying leg curl	3 or 4	8 to 12	2 minutes
Standing calf raise	3 or 4	8 to 12	2 minutes
<i>Friday (moderate upper)</i>			
Incline press	3 or 4	8 to 12	2 minutes
Flat dumbbell fly	3 or 4	8 to 12	2 minutes
Lat pulldown	3 or 4	8 to 12	2 minutes
One-arm dumbbell row	3 or 4	8 to 12	2 minutes
Military press	3 or 4	8 to 12	2 minutes
Machine lateral raise	3 or 4	8 to 12	2 minutes
Cable abdominal crunch	3 or 4	8 to 12	2 minutes
Week 2			
<i>Monday (light lower)</i>			
Dumbbell lunge	2 or 3	15 to 25	30 to 60 seconds
Knee extension	2 or 3	15 to 25	30 to 60 seconds
Cable glute hip extension	2 or 3	15 to 25	30 to 60 seconds
Seated leg curl	2 or 3	15 to 25	30 to 60 seconds
Reverse hyperextension	2 or 3	15 to 25	30 to 60 seconds
Seated calf raise	2 or 3	15 to 25	30 to 60 seconds
<i>Tuesday (light upper)</i>			
Hammer chest press	2 or 3	15 to 25	30 to 60 seconds
Cable fly	2 or 3	15 to 25	30 to 60 seconds
Cross cable pulldown	2 or 3	15 to 25	30 to 60 seconds
Seated pulley row	2 or 3	15 to 25	30 to 60 seconds
Dumbbell seated shoulder press	2 or 3	15 to 25	30 to 60 seconds
Dumbbell rear deltoid raise	2 or 3	15 to 25	30 to 60 seconds
Reverse crunch	2 or 3	15 to 25	30 to 60 seconds

Concepts adapted from B.J. Schoenfeld, 2013, *The M.A.X. Muscle Plan* (Champaign, IL: Human Kinetics, 2013).

TABLE 8.6 Sample Modified Linear Periodized Program for Loading

Exercise	Sets	Repetitions	Rest interval
Strength phase			
Microcycle 1: total-body program, 3 weeks of training 3 days per week			
Monday, Wednesday, Friday			
Bench press	3	3 to 5	3 minutes
Barbell bent reverse row	3	3 to 5	3 minutes
Standing military press	3	3 to 5	3 minutes
Barbell squat	3	3 to 5	3 minutes
Deadlift	3	3 to 5	3 minutes
Microcycle 2 (deload): 1 week of training 2 days per week			
Monday, Thursday			
Incline chest fly	3	15 to 20	2 to 3 minutes
Front lat pulldown	3	15 to 20	2 to 3 minutes
Barbell upright row	3	15 to 20	2 to 3 minutes
Bulgarian squat	3	15 to 20	2 to 3 minutes
Lying hamstring curl	3	15 to 20	2 to 3 minutes
Standing calf raise	3	15 to 20	2 to 3 minutes
Microcycle 3: upper/lower split body, 3 weeks of training 4 days per week			
Monday, Thursday			
Barbell chest press	4 or 5	3 to 5	3 minutes
Incline dumbbell fly press	3	6 to 8	2 minutes
Barbell reverse row	4 or 5	3 to 5	3 minutes
Lat pulldown	3	6 to 8	2 minutes
Standing military press	4 or 5	3 to 5	3 minutes
Dumbbell lateral raise	3	6 to 8	2 minutes
Tuesday, Friday			
Squat	4 or 5	3 to 5	3 minutes
Deadlift	4 or 5	3 to 5	3 minutes
Good morning	3	6 to 8	2 minutes
Lying hamstring curl	3	6 to 8	2 minutes
Standing calf raise	3	6 to 8	2 minutes
Metabolic phase			
Microcycle 1: total-body program, 3 weeks of training 3 days per week			
Monday, Wednesday, Friday			
Incline dumbbell chest press	3	15 to 25	30 to 60 seconds
One-arm dumbbell row	3	15 to 25	30 to 60 seconds
Dumbbell shoulder press	3	15 to 25	30 to 60 seconds
Seated dumbbell curl	3	15 to 25	30 to 60 seconds
Dumbbell overhead triceps extension	3	15 to 25	30 to 60 seconds
Leg press	3	15 to 25	30 to 60 seconds
Lying hamstring curl	3	15 to 25	30 to 60 seconds
Standing calf raise	3	15 to 25	30 to 60 seconds

(continued)

Table 8.6 (continued)

Exercise	Sets	Repetitions	Rest interval
Metabolic phase			
Microcycle 2 (deload): 1 week of training 2 days per week			
Monday, Thursday			
Incline chest fly	3	15 to 20	2 to 3 minutes
Front lat pulldown	3	15 to 20	2 to 3 minutes
Barbell upright row	3	15 to 20	2 to 3 minutes
Bulgarian squat	3	15 to 20	2 to 3 minutes
Lying hamstring curl	3	15 to 20	2 to 3 minutes
Standing calf raise	3	15 to 20	2 to 3 minutes
Hypertrophy phase			
Microcycle 1: total-body program, 3 weeks of training 3 days per week			
Monday			
Dumbbell chest press	3 or 4	6 to 12	2 minutes
Seated pulley row	3 or 4	6 to 12	2 minutes
Military press	3 or 4	6 to 12	2 minutes
Incline dumbbell curl	2 or 3	6 to 12	2 minutes
Triceps pushdown	2 or 3	6 to 12	2 minutes
Front squat	3 or 4	6 to 12	2 minutes
Seated hamstring curl	3 or 4	6 to 12	2 minutes
Standing calf raise	3 or 4	6 to 12	2 minutes
Wednesday			
Incline barbell chest press	3 or 4	6 to 12	2 minutes
Lat pulldown	3 or 4	6 to 12	2 minutes
Cable lateral raise	3 or 4	6 to 12	2 minutes
Hammer curl	2 or 3	6 to 12	2 minutes
Lying triceps extension	2 or 3	6 to 12	2 minutes
Hack squat	3 or 4	6 to 12	2 minutes
Romanian deadlift	3 or 4	6 to 12	2 minutes
Seated calf raise	3 or 4	6 to 12	2 minutes
Friday			
Cable chest fly	3 or 4	6 to 12	2 minutes
One-arm dumbbell row	3 or 4	6 to 12	2 minutes
Rear delt raise	3 or 4	6 to 12	2 minutes
EZ curl	2 or 3	6 to 12	2 minutes
Overhead triceps extension	2 or 3	6 to 12	2 minutes
Leg press	3 or 4	6 to 12	2 minutes
Lying leg curl	3 or 4	6 to 12	2 minutes
Toe press	3 or 4	6 to 12	2 minutes
Microcycle 2 (deload): 1 week of training 2 days per week			
Monday, Thursday			
Incline chest fly	3	15 to 20	2 to 3 minutes
Front lat pulldown	3	15 to 20	2 to 3 minutes
Barbell upright row	3	15 to 20	2 to 3 minutes
Bulgarian squat	3	15 to 20	2 to 3 minutes
Lying hamstring curl	3	15 to 20	2 to 3 minutes
Standing calf raise	3	15 to 20	2 to 3 minutes

Exercise	Sets	Repetitions	Rest interval
Hypertrophy phase			
Microcycle 3: upper/lower split body, 3 weeks of training 4 days per week			
Monday			
Barbell flat press	3 or 4	6 to 12	2 minutes
Incline dumbbell fly	3 or 4	6 to 12	2 minutes
Reverse lat pulldown	3 or 4	6 to 12	2 minutes
Seated wide-grip cable row	3 or 4	6 to 12	2 minutes
Dumbbell shoulder press	3 or 4	6 to 12	2 minutes
Cable lateral raise	3 or 4	6 to 12	2 minutes
Barbell curl	3 or 4	6 to 12	2 minutes
Overhead dumbbell triceps extension	3 or 4	6 to 12	2 minutes
Tuesday			
Barbell split squat	3 or 4	6 to 12	2 minutes
Knee extension	3 or 4	6 to 12	2 minutes
Stiff-legged deadlift	3 or 4	6 to 12	2 minutes
Lying leg curl	3 or 4	6 to 12	2 minutes
Standing calf raise	3 or 4	6 to 12	2 minutes
Seated calf raise	3 or 4	6 to 12	2 minutes
Cable kneeling twisting rope crunch	3 or 4	6 to 12	2 minutes
Thursday			
Incline machine press	3 or 4	6 to 12	2 minutes
Pec deck	3 or 4	6 to 12	2 minutes
Chin-up	3 or 4	6 to 12	2 minutes
One-arm dumbbell row	3 or 4	6 to 12	2 minutes
Dumbbell shoulder press	3 or 4	6 to 12	2 minutes
Kneeling cable reverse fly	3 or 4	6 to 12	2 minutes
Dumbbell biceps curl	2 or 3	6 to 12	2 minutes
Dumbbell triceps kickback	2 or 3	6 to 12	2 minutes
Friday			
Leg press	3 or 4	6 to 12	2 minutes
Dumbbell side lunge	3 or 4	6 to 12	2 minutes
Hyperextension	3 or 4	6 to 12	2 minutes
Seated leg curl	3 or 4	6 to 12	2 minutes
Seated calf raise	3 or 4	6 to 12	2 minutes
Toe press	3 or 4	6 to 12	2 minutes
Reverse crunch	3 or 4	6 to 12	2 minutes
Microcycle 4 (deload): 1 week of training 2 days per week			
Monday, Thursday			
Incline chest fly	3	15 to 20	2 to 3 minutes
Front lat pulldown	3	15 to 20	2 to 3 minutes
Barbell upright row	3	15 to 20	2 to 3 minutes
Bulgarian squat	3	15 to 20	2 to 3 minutes
Lying hamstring curl	3	15 to 20	2 to 3 minutes
Standing calf raise	3	15 to 20	2 to 3 minutes

(continued)

Table 8.6 (continued)

Exercise	Sets	Repetitions	Rest interval
Hypertrophy phase			
Microcycle 5: 3-way split body, 3 weeks of training 6 days per week			
Monday, Friday			
Lat pulldown	3 or 4	6 to 12	2 minutes
One-arm dumbbell row	3 or 4	6 to 12	2 minutes
Dumbbell pullover	3 or 4	6 to 12	2 minutes
Incline barbell press	3 or 4	6 to 12	2 minutes
Decline dumbbell press	3 or 4	6 to 12	2 minutes
Cable fly	3 or 4	6 to 12	2 minutes
Barbell abdominal rollout	3 or 4	6 to 12	2 minutes
Twisting crunch	3 or 4	6 to 12	2 minutes
Tuesday, Saturday			
Barbell back squat	3 or 4	6 to 12	2 minutes
Dumbbell lunge	3 or 4	6 to 12	2 minutes
Knee extension	3 or 4	6 to 12	2 minutes
Hip thrust	3 or 4	6 to 12	2 minutes
Barbell stiff-legged deadlift	3 or 4	6 to 12	2 minutes
Leg curl	3 or 4	6 to 12	2 minutes
Standing calf raise	3 or 4	6 to 12	2 minutes
Seated calf raise	3 or 4	6 to 12	2 minutes
Wednesday, Sunday			
Barbell military press	3 or 4	6 to 12	2 minutes
Machine lateral raise	3 or 4	6 to 12	2 minutes
Machine rear delt fly	3 or 4	6 to 12	2 minutes
Cable overhead triceps extension	2 or 3	6 to 12	2 minutes
Hammer curl	2 or 3	6 to 12	2 minutes
Lying triceps extension	2 or 3	6 to 12	2 minutes
Concentration curl	2 or 3	6 to 12	2 minutes
Cable triceps kickback	2 or 3	6 to 12	2 minutes
Dumbbell incline curl	2 or 3	6 to 12	2 minutes
Microcycle 6 (active recovery): 1 week of light recreational activity only			

Concepts adapted from B.J. Schoenfeld, 2013, *The M.A.X. Muscle Plan* (Champaign, IL: Human Kinetics, 2013).

in a recent study on German volume training (60). Recreationally trained men were assigned to perform a 12-week split-body routine with either 5 or 10 sets performed per exercise. Results showed greater increases in lean soft-tissue mass in the legs in the higher-volume condition than in the lower-volume condition after the initial 6-week training period (1 kg, or 2.2 lb, versus no gain, respectively). However, results regressed over the final 6 weeks of the study, and those training with higher volumes lost all of their acquired gains. Although speculative, this sug-

gests the musculature had become desensitized to the volume stimulus with corresponding negative effects to the neuroendocrine system.

It is important to consider volume programming in terms of the overall number of sets performed for all muscle groups over a given time period (e.g., weekly). Overtraining is a systemic phenomenon brought about by overtaxing bodily systems (e.g., neuromuscular, endocrinological, immunological) as a whole. Thus, each person has a certain volume he or she can perform over time without incurring

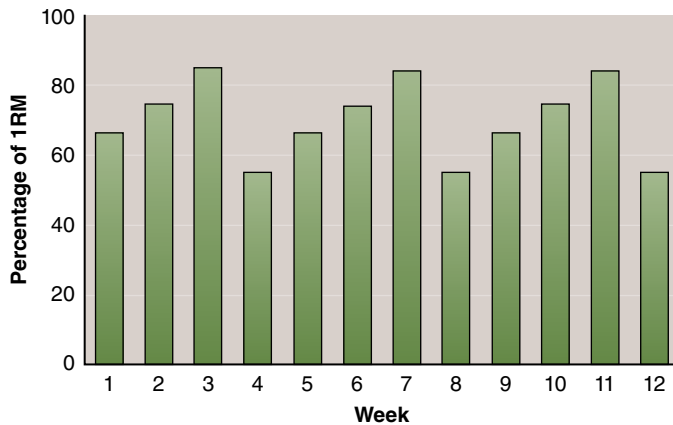


FIGURE 8.7 The wavelike loading pattern of step loading in a hypertrophy mesocycle.

negative consequences. The concept can be likened to a budget: The amount of money available is fixed, and when an item is purchased, it reduces the amount left to spend on other items. Similarly, when adding volume for a given muscle group, volume for another muscle group should be proportionally decreased so that training volume for the body as a whole remains relatively constant (i.e., within the volume budget). It therefore makes sense to “save up” volume to “spend” on lagging muscle groups and correspondingly perform fewer sets for muscle groups that respond well to training.

While this strategy provides a viable means for guiding volume prescription, it simplifies a topic that is nuanced. In particular, it disregards the fact that the specific components of an exercise also affect fatigue of bodily systems. These components include the modality (free weights versus machines), body region (upper versus lower), and number of joints involved in performance. For example, performing multiple sets of squats is substantially more taxing, both from a neuromuscular and physiological (i.e., metabolic) standpoint, than performing a similar number of sets of arm curls; thus, in comparison to squats, a greater volume of arm curls could be included in a routine without initiating an overtrained response. Hence, considerations specific to the exercises employed also must be factored into decision making when determining volume and distribution over a given training cycle.

Maximum recoverable volume (MRV) has been proposed as a strategy to manage volume across a training cycle (66). MRV is a performance-oriented concept that assesses recovery based on an individual’s ability to maintain loading over time. It can be operationally defined as the maximal number of sets that an individual can perform in a given unit of time and still recover from in that time frame, usually defined as a week or the length of a microcycle. If the loads used increase from the previous cycle or remain stable, then the volume of the previous microcycle was lower than MRV and volume can be added to the program. Alternatively, if the amount lifted is less than in the previous microcycle, the person may have exceeded his or her MRV and would likely benefit from reducing volume and thus fatigue in the next microcycle. Although no direct research has been conducted on the topic, intuitively there appears to be a rationale for using MRV to help guide volume prescription.

The hypertrophy phase in table 8.6 illustrates a strategy for systematically increasing volume across a training cycle. This strategy can be used in both linear and undulating models. Microcycle 1 shows a 3-day-per-week routine in which all major muscles are trained in each workout session. In this scheme, training would generally be carried out on nonconsecutive days (e.g., Mondays, Wednesdays, and Fridays); the other days are reserved for recovery. Microcycle 3 increases frequency to 4 days per week employing an upper-body/lower-body split routine. This type of routine is often carried out on a 2-on/1-off, 2-on/2-off basis (e.g., training on Mondays, Tuesdays, Thursdays, and Fridays). Although training volume remains the same on a per-session basis, total weekly volume is greater because of the higher frequency of training. Microcycle 5 increases frequency to 6 days per week employing a traditional body-building-style split routine. Typically training in this type of protocol is carried out on a 3-on/1-off basis (e.g., training on Monday, Tuesday, Wednesday, Friday, Saturday, and Sunday). Again, the per-session training volume remains constant, as with the previous protocols, but weekly volume is further increased as a result of more frequent training.

Periodizing Exercise Selection

As previously noted, maximizing hypertrophy requires performing a variety of exercises that work the musculature from different angles and in different planes of motion, while taking into account biomechanical and physiological factors. There are many ways to accomplish this task. One consideration is the frequency of variation of exercise selection. Some popular fitness programs advocate changing up exercises on a session-by-session basis, based on the premise that “muscle confusion” is a key to optimizing results (29). However, the concept of muscle confusion is not supported in the literature, and there is evidence that randomly rotating exercises via a computerized app does not enhance hypertrophic adaptations and may in fact blunt muscular development (10). Although the reasons are not clear, it is possible that changing exercises too frequently may impede performance, which in turn can reduce the ability to exert maximal tension to the target muscles.

Research into the effects of periodizing exercise selection is scant. A recent study indi-

cated that autoregulated exercise selection, whereby individuals self-selected the exercises for each training session, produced modestly greater increases in lean mass compared to a fixed exercise protocol (106). This provides evidence that allowing exercise choice in a training program can be beneficial, perhaps by enhancing motivation to train or affording the ability to tailor movements to individual preferences, or both.

Although the dearth of scientific evidence precludes the ability to offer definitive recommendations on the topic, a case can be made to keep more complex exercises in a regular rotation because they require consistent practice in order to maintain optimal performance. Multi-joint, free-weight exercises such as squats, rows, and presses are the most suitable. Alternatively, movements that involve reduced degrees of freedom (e.g., machine-based, single-joint exercises) can be varied more liberally in order to provide greater novelty and thereby possibly enhance muscular development.

TAKE-HOME POINTS

- Several biomechanical considerations need to be taken into account when selecting exercises for a hypertrophy-oriented program. These include length–tension relationship, training angle, plane of movement, spacing of hands and feet, and exercise type.
- The application of biomechanical principles to exercise selection is specific to a given muscle, its architecture, and the joint at which it originates. Combining exercises based on applied anatomy and kinesiology is essential to ensuring the complete development of the major musculature.
- Hypertrophy-oriented training programs should be periodized to promote continued gains while reducing the risk of overtraining. Several periodized models can be employed to maximize muscle mass, including linear, undulating, and reverse linear approaches. Research has not shown one model to be superior over another, and each can thus be considered a viable strategy in program design. Importantly, periodization is a general concept, not a rigid training system; therefore, the implementation of the models should be adapted based on the needs and abilities of the lifter.
- Deload periods that reduce intensity or volume, or both, should be integrated into periodized programs to facilitate rejuvenation and recovery. A 3:1 ratio (in weeks) of training and deloading is a good guideline to use as a starting point. Modifications should then be made depending on individual response.